

ASSIGNMENT 4: RESPONSIVE DASHBOARD UI

[Github Link](#)

AIM:

To create a responsive multi-section dashboard in Flutter using ListView, GridView, MediaQuery, and Flexible/Expanded widgets that adapts to different screen sizes.

OBJECTIVE:

- To understand responsive UI design in Flutter.
- To use MediaQuery for detecting screen dimensions.
- To use GridView for arranging dashboard statistics.
- To use ListView for displaying scrollable content.
- To use Flexible and Expanded for proper space distribution.
- To create reusable widgets and organize the application into multiple files.

IMPLEMENTATION:

A modern student dashboard named "StudyFlow" was developed. The application was divided into multiple Dart files instead of keeping all the code inside main.dart.

PROJECT STRUCTURE:

- main.dart - Application entry point.
- app.dart - Application configuration and theme.
- models/dashboard_data.dart - Dashboard data and model.
- screens/dashboard_screen.dart - Main dashboard screen.
- screens/analytics_screen.dart - Analytics section.
- screens/projects_screen.dart - Projects section.
- screens/tasks_screen.dart - Task management section.
- widgets/sidebar.dart - Responsive navigation sidebar.
- widgets/stat_card.dart - Reusable statistics card.
- widgets/overview_card.dart - Learning and coding overview cards.
- widgets/activity_tile.dart - Recent activity item.

MAIN FEATURES:

1. Responsive Navigation
 - Sidebar is displayed on larger screens.
 - Drawer/navigation is used on smaller screens.
 - Navigation allows switching between Dashboard, Analytics, Projects and Tasks.
2. Dashboard Overview
 - Displays total projects, tasks, completed tasks and progress.
 - Uses reusable statistic cards.
 - Displays learning and coding progress.
3. Analytics
 - Provides a separate section for viewing learning and progress information.

4. Projects

- Displays project-related information in a responsive layout.

5. Tasks

- Displays task progress and task items.
- Supports task filtering and interactions.
- Allows users to mark tasks as completed and add tasks.

6. Recent Activity

- Displays recent learning and development activities using reusable activity tiles.

7. Theme Support

- Supports light and dark themes.

FLUTTER WIDGETS AND CONCEPTS USED:

- Scaffold
- Row
- Column
- Container
- ListView
- GridView
- MediaQuery
- Flexible
- Expanded
- Card
- Text
- Icon
- IconButton
- ChoiceChip
- Drawer
- Navigation
- LinearProgressIndicator

RESPONSIVE DESIGN:

MediaQuery is used to determine the available screen width and adjust the interface accordingly. GridView changes the number of columns according to the available screen size. Flexible and Expanded are used to distribute available space and prevent layout overflow.

TESTING:

The application was tested for:

- Dashboard navigation.
- Analytics, Projects and Tasks screens.
- Responsive layout.
- Sidebar and mobile navigation.
- Task filtering and interactions.

- Theme switching.
- Scrolling through dashboard content.
- Flutter code analysis.

The project was analyzed using:

flutter analyze

Result:

No issues found.

RESULT:

A responsive and interactive multi-section Student Dashboard was successfully created using Flutter. The application demonstrates responsive layouts, reusable widgets, navigation, dynamic interactions and organized multi-file architecture.

CONCLUSION:

This assignment provided practical experience in building responsive Flutter applications. It improved understanding of ListView, GridView, MediaQuery, Flexible, Expanded, navigation and reusable widgets. The final application is organized, responsive, interactive and suitable for different screen sizes.